

%Important to note that this program takes a fair amount of time to finish.
%As 24 FEAs are being performed and 3 finer meshes are created.
%It takes about 2-5 mins to run

%Section 1 (Variable initialization)

load("droneArmMaterials.mat")

%To get information from file do: materials(i).(TopRowName)

%Ex. materials(1).name will give Carbon Fiber Composite (CFRP)

Max_mats = length(materials); %Used for amount of loop runs

MaxThrust = 4; %(kg)

DroneWeightAllowed = MaxThrust/2; %(kg) To ensure the 2:1 T/W ratio is followed

MinPayload = 0.5; %(kg) Min req for payload

DroneComps = 1; %(kg) Weight of other drone components

%Variables to hold and display masses later

DroneArm1 = zeros(1,Max_mats)'; %(kg)

PayloadMax1 = zeros(1,Max_mats)'; %(kg)

DroneArm2 = zeros(1,Max_mats)'; %(kg)

PayloadMax2 = zeros(1,Max_mats)'; %(kg)

DroneArm3 = zeros(1,Max_mats)'; %(kg)

PayloadMax3 = zeros(1,Max_mats)'; %(kg)

DroneArm4 = zeros(1,Max_mats)'; %(kg)

PayloadMax4 = zeros(1,Max_mats)'; %(kg)

%Importing of models. Matlab always assumes mm when handling models. Scale

%function is there to allow for proper FEA and calcs.

%models are in mm except 2

%For the best experience for importing models please use SolidWorks

gm1 = importGeometry("Triangle_ArmMod.STL");

gm2 = importGeometry("Circle_Arm.STL"); %model in m already

gm3 = importGeometry("Beam_Arm.STL");

gm4 = importGeometry("Lattice_ArmV2.STL");

gm1 = scale(gm1, 1E-3);

%scale(gm2, 1E-3);

gm3 = scale(gm3, 1E-3);

gm4 = scale(gm4, 1E-3);

md1 = femodel(AnalysisType="structuralStatic", Geometry = gm1);

md2 = femodel(AnalysisType="structuralStatic", Geometry = gm2);

md3 = femodel(AnalysisType="structuralStatic", Geometry = gm3);

```
md4 = femodel(AnalysisType="structuralStatic", Geometry = gm4);
```

```
%No longer needed. Can be accessed through mds
```

```
clearvars gm1 gm2 gm3 gm4
```

```
%model 2 required are coarser mesh
```

```
%model 3 required a finer one compared to the others.
```

```
md1 = generateMesh(md1, Hmin=.0005, Hmax=1);
```

```
md2 = generateMesh(md2);
```

```
md3 = generateMesh(md3, Hmin=.00005, Hmax=1);
```

```
md4 = generateMesh(md4, Hmin=.0005, Hmax=1);
```

```
%Section 2 (For devs)
```

```
%This section is for other programmers adapting the code for their models.
```

```
%These pde functions are here to allow you to quickly run this section to
```

```
%see where the faces are for the model
```

```
%figure
%pdemesh(md1);
% %figure
%pdemesh(md2);
% %figure
% pdemesh(md3);
%
% pdemesh(md4);
%
%
% figure
% pdegplot(md1,FaceLabels="on", FaceAlpha=.3);
% figure
% pdegplot(md1,FaceLabels="on", FaceAlpha=1);
%
% figure
% pdegplot(md2,FaceLabels="on", FaceAlpha=0.3);
% figure
% pdegplot(md2,FaceLabels="on", FaceAlpha=1);
%
% figure
% pdegplot(md3,FaceLabels="on", FaceAlpha=0.3);
% figure
% pdegplot(md3,FaceLabels="on", FaceAlpha=1);
% figure
% pdegplot(md3,FaceLabels="off", FaceAlpha=1);
%
% figure
% pdegplot(md4,FaceLabels="on", FaceAlpha=0.3);
% figure
% pdegplot(md4,FaceLabels="on", FaceAlpha=1);
```

%

%Section 3 (Thrust to weight analysis occurs here)

names = strings(1,Max_mats)'; %Holds names of materials in a vector col for more convenient use later.

%Vol variables are created to calculate weights and costs of the drone arms
%Vol#c is created with the assumption that the material will be cut from a
%block of material. Thus using more actual material than the volume command
%of the mesh states.

% The 2 separate cost variables are the result of the
%confusion in regards to how much materials costs. cost# assumes the
%materials cost column is in (\$/m) while cost#g assumes it to be (\$/Kg)

%The vol#c calc assumes the models are aligned with the origin point and
%take up space in the positive direction.

vol1=volume(md1.Geometry.Mesh);

vol1c =

max(md1.Geometry.Vertices(:,1))*max(md1.Geometry.Vertices(:,2))*max(md1.Geometry.Vertices(:,3));

cost1 = zeros(1,Max_mats)';

cost1g = zeros(1,Max_mats)';

%Model 2 was made in a different program. So, it's vol2c was hand calculated.

vol2=volume(md2.Geometry.Mesh);

vol2c = 2.252E-5;

cost2 = zeros(1,Max_mats)';

cost2g = zeros(1,Max_mats)';

vol3=volume(md3.Geometry.Mesh);

vol3c =

max(md3.Geometry.Vertices(:,1))*max(md3.Geometry.Vertices(:,2))*max(md3.Geometry.Vertices(:,3));

cost3 = zeros(1,Max_mats)';

cost3g = zeros(1,Max_mats)';

vol4=volume(md4.Geometry.Mesh);

vol4c =

max(md4.Geometry.Vertices(:,1))*max(md4.Geometry.Vertices(:,2))*max(md4.Geometry.Vertices(:,3));

cost4 = zeros(1,Max_mats)';

cost4g = zeros(1,Max_mats)';

%Calc for drone arm weights, max payload, and costs

%This is also where the Thrust to Weight ratio (TWR) analysis occurs.

%Since the TWR is locked at 2 from the variable setup in section 1 to

%ensure the heaviest payload possible.

```
for i = 1:Max_mats
    names(i) = materials(i).name;
    DroneArm1(i) = vol1 * materials(i).rho_kg_m3;
    PayloadMax1(i) = DroneWeightAllowed - DroneArm1(i)*4 - DroneComps; %This calc
ensures a 2:1 TWR
    cost1(i) = vol1 * (materials(i).cost_USD_per_m^3);
    cost1g(i) = vol1c * materials(i).rho_kg_m3 * materials(i).cost_USD_per_m;

    DroneArm2(i) = vol2 * materials(i).rho_kg_m3;
    PayloadMax2(i) = DroneWeightAllowed - DroneArm2(i)*4 - DroneComps;
    cost2(i) = vol2c * (materials(i).cost_USD_per_m)^3;
    cost2g(i) = vol2c * materials(i).rho_kg_m3* materials(i).cost_USD_per_m;

    DroneArm3(i) = vol3 * materials(i).rho_kg_m3;
    PayloadMax3(i) = DroneWeightAllowed - DroneArm3(i)*4 - DroneComps;
    cost3(i) = vol3c * (materials(i).cost_USD_per_m)^3;
    cost3g(i) = vol3c * materials(i).rho_kg_m3* materials(i).cost_USD_per_m;

    DroneArm4(i) = vol4 * materials(i).rho_kg_m3;
    PayloadMax4(i) = DroneWeightAllowed - DroneArm4(i)*4 - DroneComps;
    cost4(i) = vol4c * (materials(i).cost_USD_per_m)^3;
    cost4g(i) = vol4c * materials(i).rho_kg_m3* materials(i).cost_USD_per_m;

end
```

%Section 4 (Applying locking faces and loading forces)

%This section is where the restraints and loads are placed on the model
%Viewing the models will show that some of the models are orientated
%differently. Which is why the loads are either applied on the y or z axis

%9.07 is the force that area of the arm faces.

% 9.8 - (9.8 * motor weight(65g)* propeller weight(10g)) = 9.065N

%It was then rounded up to 9.07N.

%It is then divided by the surface area of the face it is attached to ($2\pi r \cdot \text{depth}$)

%Which makes the results give an approximation for FEAs

%Lastly sanity checks were done hand calced and using SolidWorks as a

%reference for results.

```
md1.FaceBC([63,64,65,66]) = faceBC(Constraint="fixed");
md1.FaceLoad(58) = faceLoad(SurfaceTraction= [0;0;9.07/(2*pi*7.5*6*1e-6)]);
md1.CellLoad = cellLoad(Gravity=[0 0 -9.8]);

md2.FaceBC([12]) = faceBC(Constraint="fixed");
md2.FaceLoad(11) = faceLoad(SurfaceTraction= [0; 9.07/(2*pi*3.5*5*1e-6); 0]);
```

```

md2.CellLoad = cellLoad(Gravity=[0, -9.8, 0]);

md3.FaceBC([12,13,14,15]) = faceBC(Constraint="fixed");
md3.FaceLoad(11) = faceLoad(SurfaceTraction= [0;0;9.07/(2*pi*1.5*4.8*1e-6)]);
md3.CellLoad = cellLoad(Gravity=[0 0 -9.8]);

md4.FaceBC([3,24,25,34]) = faceBC(Constraint="fixed");
md4.FaceLoad(26) = faceLoad(SurfaceTraction= [0;9.07/(2*pi*2.65*1.9*1e-6);0]);
md4.CellLoad = cellLoad(Gravity=[0 -9.8 0]);

```

%Section 5 (FEA calcs)

%Pass this point are 4 big sections of code where FEA is performed on the 4
 %drone arms. They are handled via switch case. Separate variables are used
 %to store the FEA results in order for the built in "Visualize PDE Results"
 %to allow the results to be selectable and viewable.

%Explanation comments return after this section

%Model 1

```

Max_Stress1 = zeros(1,6)';
Max_Displ1 = zeros(1,6)';
FoS1 = zeros(1,6)';
for i = 1:Max_mats

    md1.MaterialProperties = ...
        materialProperties(YoungsModulus=materials(i).E_Pa, ...
        PoissonsRatio=materials(i).nu, ...
        MassDensity=materials(i).rho_kg_m3);

    switch i
        case 1
            Model_1_result_CFRP = solve(md1);
            Max_Stress1(i) = max(Model_1_result_CFRP.VonMisesStress);
            Max_Displ1(i) = max(Model_1_result_CFRP.Displacement.Magnitude);
            FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);
        case 2
            Model_1_result_Al_Alloy = solve(md1);
            Max_Stress1(i) = max(Model_1_result_Al_Alloy.VonMisesStress);
            Max_Displ1(i) = max(Model_1_result_Al_Alloy.Displacement.Magnitude);
            FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);
        case 3
            Model_1_result_GFRP = solve(md1);
            Max_Stress1(i) = max( Model_1_result_GFRP.VonMisesStress);
            Max_Displ1(i) = max( Model_1_result_GFRP.Displacement.Magnitude);
            FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);
        case 4
            Model_1_result_PLA = solve(md1);
            Max_Stress1(i) = max(Model_1_result_PLA.VonMisesStress);

```

```

        Max_Displ1(i) = max(Model_1_result_PLA.Displacement.Magnitude);
        FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);
    case 5
        Model_1_result_ASB = solve(md1);
        Max_Stress1(i) = max(Model_1_result_ASB.VonMisesStress);
        Max_Displ1(i) = max(Model_1_result_ASB.Displacement.Magnitude);
        FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);
    case 6
        Model_1_result_Wood = solve(md1);
        Max_Stress1(i) = max(Model_1_result_Wood.VonMisesStress);
        Max_Displ1(i) = max(Model_1_result_Wood.Displacement.Magnitude);
        FoS1(i) = materials(i).yieldStrength_Pa / Max_Stress1(i);

end

end

```

```

%For model 2
Max_Stress2 = zeros(1,6)';
Max_Displ2 = zeros(1,6)';
FoS2 = zeros(1,6)';
for i = 1:Max_mats

    md2.MaterialProperties = ...
        materialProperties(YoungsModulus=materials(i).E_Pa, ...
        PoissonsRatio=materials(i).nu, ...
        MassDensity=materials(i).rho_kg_m3);

    switch i
        case 1
            Model_2_result_CFRP = solve(md2);
            Max_Stress2(i) = max(Model_2_result_CFRP.VonMisesStress);
            Max_Displ2(i) = max(Model_2_result_CFRP.Displacement.Magnitude);
            FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);
        case 2
            Model_2_result_Al_Alloy = solve(md2);
            Max_Stress2(i) = max(Model_2_result_Al_Alloy.VonMisesStress);
            Max_Displ2(i) = max(Model_2_result_Al_Alloy.Displacement.Magnitude);
            FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);
        case 3
            Model_2_result_GFRP = solve(md2);
            Max_Stress2(i) = max( Model_2_result_GFRP.VonMisesStress);
            Max_Displ2(i) = max( Model_2_result_GFRP.Displacement.Magnitude);
            FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);
        case 4
            Model_2_result_PLA = solve(md2);

```

```

        Max_Stress2(i) = max(Model_2_result_PLA.VonMisesStress);
        Max_Disp2(i) = max(Model_2_result_PLA.Displacement.Magnitude);
        FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);
    case 5
        Model_2_result_ASB = solve(md2);
        Max_Stress2(i) = max(Model_2_result_ASB.VonMisesStress);
        Max_Disp2(i) = max(Model_2_result_ASB.Displacement.Magnitude);
        FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);
    case 6
        Model_2_result_Wood = solve(md2);
        Max_Stress2(i) = max(Model_2_result_Wood.VonMisesStress);
        Max_Disp2(i) = max(Model_2_result_Wood.Displacement.Magnitude);
        FoS2(i) = materials(i).yieldStrength_Pa / Max_Stress2(i);

end

end
end

```

```

%For model 3

```

```

Max_Stress3 = zeros(1,6)';
Max_Disp3 = zeros(1,6)';
FoS3 = zeros(1,6)';
for i = 1:Max_mats

    md3.MaterialProperties = ...
        materialProperties(YoungsModulus=materials(i).E_Pa, ...
        PoissonsRatio=materials(i).nu, ...
        MassDensity=materials(i).rho_kg_m3);

    switch i
        case 1
            Model_3_result_CFRP = solve(md3);
            Max_Stress3(i) = max(Model_3_result_CFRP.VonMisesStress);
            Max_Disp3(i) = max(Model_3_result_CFRP.Displacement.Magnitude);
            FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);
        case 2
            Model_3_result_Al_Alloy = solve(md3);
            Max_Stress3(i) = max(Model_3_result_Al_Alloy.VonMisesStress);
            Max_Disp3(i) = max(Model_3_result_Al_Alloy.Displacement.Magnitude);
            FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);
        case 3
            Model_3_result_GFRP = solve(md3);
            Max_Stress3(i) = max( Model_3_result_GFRP.VonMisesStress);
            Max_Disp3(i) = max( Model_3_result_GFRP.Displacement.Magnitude);
            FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);
        case 4
            Model_3_result_PLA = solve(md3);

```

```

        Max_Stress3(i) = max(Model_3_result_PLA.VonMisesStress);
        Max_Disp3(i) = max(Model_3_result_PLA.Displacement.Magnitude);
        FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);
    case 5
        Model_3_result_ASB = solve(md3);
        Max_Stress3(i) = max(Model_3_result_ASB.VonMisesStress);
        Max_Disp3(i) = max(Model_3_result_ASB.Displacement.Magnitude);
        FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);
    case 6
        Model_3_result_Wood = solve(md3);
        Max_Stress3(i) = max(Model_3_result_Wood.VonMisesStress);
        Max_Disp3(i) = max(Model_3_result_Wood.Displacement.Magnitude);
        FoS3(i) = materials(i).yieldStrength_Pa / Max_Stress3(i);

end

end
end

```

```

%For model 4
Max_Stress4 = zeros(1,6)';
Max_Disp4 = zeros(1,6)';
FoS4 = zeros(1,6)';
for i = 1:Max_mats

    md4.MaterialProperties = ...
        materialProperties(YoungsModulus=materials(i).E_Pa, ...
        PoissonsRatio=materials(i).nu, ...
        MassDensity=materials(i).rho_kg_m3);

    switch i
        case 1
            Model_4_result_CFRP = solve(md4);
            Max_Stress4(i) = max(Model_4_result_CFRP.VonMisesStress);
            Max_Disp4(i) = max(Model_4_result_CFRP.Displacement.Magnitude);
            FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);
        case 2
            Model_4_result_Al_Alloy = solve(md4);
            Max_Stress4(i) = max(Model_4_result_Al_Alloy.VonMisesStress);
            Max_Disp4(i) = max(Model_4_result_Al_Alloy.Displacement.Magnitude);
            FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);
        case 3
            Model_4_result_GFRP = solve(md4);
            Max_Stress4(i) = max( Model_4_result_GFRP.VonMisesStress);
            Max_Disp4(i) = max( Model_4_result_GFRP.Displacement.Magnitude);
            FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);
        case 4

```



```

        Model_4_result_PLA = solve(md4);
        Max_Stress4(i) = max(Model_4_result_PLA.VonMisesStress);
        Max_Disp4(i) = max(Model_4_result_PLA.Displacement.Magnitude);
        FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);
    case 5
        Model_4_result_ASB = solve(md4);
        Max_Stress4(i) = max(Model_4_result_ASB.VonMisesStress);
        Max_Disp4(i) = max(Model_4_result_ASB.Displacement.Magnitude);
        FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);
    case 6
        Model_4_result_Wood = solve(md4);
        Max_Stress4(i) = max(Model_4_result_Wood.VonMisesStress);
        Max_Disp4(i) = max(Model_4_result_Wood.Displacement.Magnitude);
        FoS4(i) = materials(i).yieldStrength_Pa / Max_Stress4(i);

end

end
end

```

%Section 6 (Checks for arm viability)

```

Useable1 = strings(1,Max_mats)';
Useable2 = strings(1,Max_mats)';
Useable3 = strings(1,Max_mats)';
Useable4 = strings(1,Max_mats)';

%Performs a check and stores it in a variable to display later. Allowing
%for a quick check way to ensure that the arm fits the criteria in regards to
%Factor of safety (FoS), can handle a 0.5Kg payload, and a Displacement of
%less than 1mm
for i=1:Max_mats
    if PayloadMax1(i) >= MinPayLoad && FoS1(i) >= 1.5 && Max_Disp1(i) <= 0.001
        Useable1(i) = 'Yes';
    else
        Useable1(i) = 'No';
    end
    if PayloadMax2(i) >= MinPayLoad && FoS2(i) >= 1.5 && Max_Disp2(i) <= 0.001
        Useable2(i) = 'Yes';
    else
        Useable2(i) = 'No';
    end

    if PayloadMax3(i) >= MinPayLoad && FoS3(i) >= 1.5 && Max_Disp3(i) <= 0.001
        Useable3(i) = 'Yes';
    else
        Useable3(i) = 'No';
    end
end

```

```

    if PayloadMax4(i) >= MinPayLoad && FoS4(i) >= 1.5 && Max_Disp4(i) <= 0.001
        Useable4(i) = 'Yes';
    else
        Useable4(i) = 'No';
    end
end
end

```

%Section 7 (Create the tables to be displayed)

%Creates tables to display the results
 %The table is done this way to allow for
 %quicker table modifications.

%All tables follow the same structure so, refer to Show1 for what each
 %column means.

TWR = [2;2;2;2;2;2]; %The program assumes maximization of payload weight.
 % So, thrust to weight ratio (TWR) will always be a 2:1 ratio.

```

Show1 = table();
Show1.Material = names; %Display the material of what the arm is made of
Show1.Mass_of_Arm_kg=DroneArm1; %Displays the weight of one drone arm with the
given material
Show1.Max_Payload_kg = PayloadMax1; %Displays the max payload capacity with the
given material for the arm
Show1.Acceptable = Useable1; %Displays whether the arm meets the minimum
requirements with the given material
Show1.Max_Disp_mm = Max_Disp1*1E3; %Max displacement of the arm with the given
material
Show1.Max_Stress_MPa = Max_Stress1*1E-6; %Max stress of the arm with the given
material
Show1.Factor_of_Safety = FoS1; %Factor of safety of the arm with the given material
Show1.Thrust_to_Weight_Ratio = TWR; %Makes a column for the table to display TWR
"Results"
%Show1.Cost = cost1;
Show1.Total_Cost_Vol_USD = cost1*4; %Cost of 4 arms with the given material using
($/m)
Show1.Total_Cost_mass_USD = cost1g*4; %Cost of 4 arms with the given material using
($/Kg)

Show2 = table();
Show2.Material = names;
Show2.Mass_of_Arm_kg=DroneArm2;
Show2.Max_Payload_kg = PayloadMax2;
Show2.Acceptable = Useable2;
Show2.Max_Disp_mm = Max_Disp2*1E3;
Show2.Max_Stress_MPa = Max_Stress2*1E-6;
Show2.Factor_of_Safety = FoS2;
Show2.Thrust_to_Weight_Ratio = TWR;

```

```

>Show2.Cost = cost2;
Show2.Total_Cost_Vol_USD = cost2*4;
Show2.Total_Cost_mass_USD = cost2g*4;

Show3 = table();
Show3.Material = names;
Show3.Mass_of_Arm_kg=DroneArm3;
Show3.Max_Payload_kg = PayloadMax3;
Show3.Acceptable = Useable3;
Show3.Max_Disp_mm = Max_Disp3*1E3;
Show3.Max_Stress_MPa = Max_Stress3*1E-6;
Show3.Factor_of_Safety = FoS3;
Show3.Thrust_to_Weight_Ratio = TWR;
>Show3.Cost = cost3;
Show3.Total_Cost_Vol_USD = cost3*4;
Show3.Total_Cost_mass_USD = cost3g*4;

Show4 = table();
Show4.Material = names;
Show4.Mass_of_Arm_kg=DroneArm4;
Show4.Max_Payload_kg = PayloadMax4;
Show4.Acceptable = Useable4;
Show4.Max_Disp_mm = Max_Disp4*1E3;
Show4.Max_Stress_MPa = Max_Stress4*1E-6;
Show4.Factor_of_Safety = FoS4;
Show4.Thrust_to_Weight_Ratio = TWR;
>Show4.Cost = cost4;
Show4.Total_Cost_Vol_USD = cost4*4;
Show4.Total_Cost_mass_USD = cost4g*4;

```

%To display the table created above. Also to make it easier for pdf viewers
 %to read the code.
 disp('Triangle Arm / Model 1 Table');

Triangle Arm / Model 1 Table

```
disp(Show1);
```

Material	Mass_of_Arm_kg	Max_Payload_kg	Acceptable	Max_Disp_mm	Max_Stress_MPa
"Carbon Fiber Composite (CFRP)"	0.042778	0.82889	"Yes"	0.88702	24.564
"Aluminum Alloy"	0.072189	0.71125	"Yes"	0.89039	24.322
"Fiberglass Composite (GFRP)"	0.050799	0.7968	"No"	2.4754	24.412
"PLA Plastic"	0.033421	0.86632	"No"	17.814	24.866
"ABS Plastic"	0.028073	0.88771	"No"	31.233	24.893
"Wood (Birch)"	0.016042	0.93583	"No"	6.2744	25.031

```
disp('|');disp('|');disp('|'); %For spacing in the program
```

```
disp('Circle Arm / Model 2 Table');
```

Circle Arm / Model 2 Table

```
disp>Show2);
```

Material	Mass_of_Arm_kg	Max_Payload_kg	Acceptable	Max_Disp_mm	Max_Stress_MPa
"Carbon Fiber Composite (CFRP)"	0.024813	0.90075	"Yes"	0.21511	15.817
"Aluminum Alloy"	0.041873	0.83251	"Yes"	0.21558	15.094
"Fiberglass Composite (GFRP)"	0.029466	0.88214	"Yes"	0.60211	16.128
"PLA Plastic"	0.019385	0.92246	"No"	4.2826	14.703
"ABS Plastic"	0.016284	0.93486	"No"	7.5165	14.937
"Wood (Birch)"	0.009305	0.96278	"No"	1.5086	15

```
disp('|');disp('|');disp('|'); %For spacing in the program
```

```
disp('Beam Arm / Model 3 Table');
```

Beam Arm / Model 3 Table

```
disp>Show3);
```

Material	Mass_of_Arm_kg	Max_Payload_kg	Acceptable	Max_Disp_mm	Max_Stress_MPa
"Carbon Fiber Composite (CFRP)"	0.026745	0.89302	"Yes"	0.97521	56.122
"Aluminum Alloy"	0.045133	0.81947	"Yes"	0.97876	55.185
"Fiberglass Composite (GFRP)"	0.03176	0.87296	"No"	2.7284	56.411
"PLA Plastic"	0.020895	0.91642	"No"	19.455	55.193
"ABS Plastic"	0.017552	0.92979	"No"	34.129	55.42
"Wood (Birch)"	0.010029	0.95988	"No"	6.8489	55.576

```
disp('|');disp('|');disp('|'); %For spacing in the program
```

```
disp('Lattice Arm / Model 4 Table');
```

Lattice Arm / Model 4 Table

```
disp>Show4);
```

Material	Mass_of_Arm_kg	Max_Payload_kg	Acceptable	Max_Disp_mm	Max_Stress_MPa
"Carbon Fiber Composite (CFRP)"	0.091297	0.63481	"Yes"	0.057522	7.8082

"Aluminum Alloy"	0.15406	0.38375	"No"	0.056408	7.5645
"Fiberglass Composite (GFRP)"	0.10841	0.56634	"Yes"	0.15999	7.7149
"PLA Plastic"	0.071326	0.7147	"No"	1.1542	7.9334
"ABS Plastic"	0.059913	0.76035	"No"	2.0334	7.9782
"Wood (Birch)"	0.034236	0.86305	"Yes"	0.41169	8.088

%Section 8 (Visualizer)

%This is sectioned off to allow you to change results without having to
%repeat the whole code

%Since it is not properly reflected in the pdf this is the "Visualize PDE
%Results" function

%The results have the following format:

% 'Model' + number + 'result' + material name

% This should be rather self explanatory to use

%Lastly make sure Deformation is not ticked

% Data to visualize

```
meshData = Model_1_result_ASB.Mesh;
```

```
nodalData = Model_1_result_ASB.Displacement.Magnitude;
```

```
deformationData = Model_1_result_ASB.Displacement;
```

% Create PDE result visualization

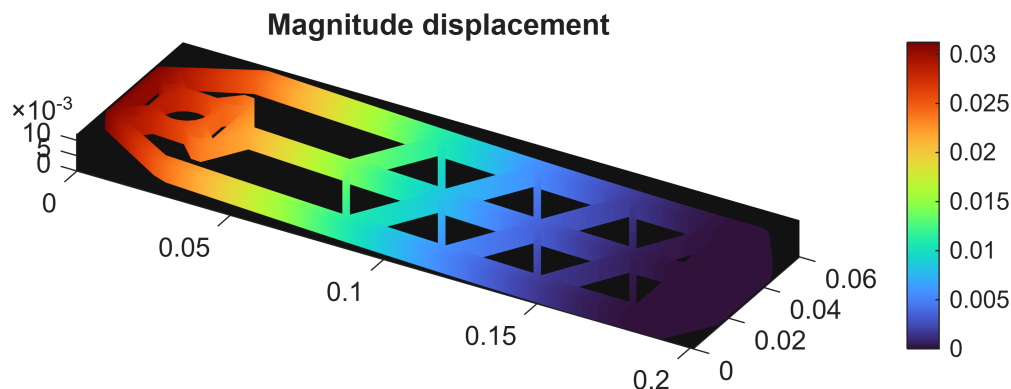
```
resultViz = pdeviz(meshData,abs(real(nodalData)), ...
```

```
    "DeformationData",deformationData, ...
```

```
    "DeformationScaleFactor",0.1921, ...
```

```
    "Title","Magnitude displacement", ...
```

```
    "ColorLimits",[0 0.03123]);
```



```
% Clear temporary variables  
clearvars meshData nodalData deformationData
```